

Strong coupling for abelian and non-abelian gauge theories: a shared existential gap

Both compact $U(1)$ and $SU(3)$ have a volume-uniform strong-coupling Kogut–Susskind gap by the Yarotsky application, with an existential threshold. The continuous-time expansion does not currently provide an explicit uniform threshold for either group. Their weak sides differ: for $SU(3)$,

$$\frac{dg^2}{d \log a} = 2b_0 g^4 + O(g^6), \quad b_0 = \frac{11}{16\pi^2} > 0,$$

whereas the continuum pure photon has no perturbative self-interaction, $b_0 = 0$. A massless weak-coupling phase of compact lattice $U(1)$ is supported by Guth (Phys. Rev. D 21 (1980) 2291) and Fröhlich–Spencer (Commun. Math. Phys. 83 (1982) 411; both metadata level). The perturbative running does not determine the whole lattice phase diagram or establish a constructive arrival in the strong-coupling region.

Correction, 2026-10-02. The earlier uniform 4λ bound, an explicit threshold of the same size as for $SU(3)$, and a target box at $g^2 \sim 4 \times 10^2$ depended on the withdrawn continuous-time count. They are withdrawn here. The finite-volume electric-loop result below survives; the volume-uniform theorem instead uses Yarotsky with an existential threshold. The earlier exact coupling-set and flow equivalences also exceeded the cited perturbative and phase results; the corrected coupling-set note separates them. The original text remains in Git history.

1. The strong-coupling region does not distinguish the groups

For compact $U(1)$ take the magnetic term $(2/g^2) \sum_p (1 - \cos \theta_p)$ and electric term $(g^2/2) \sum_\ell n_\ell^2$. The least nonzero electric eigenvalue per link is $\varepsilon = g^2/2$. The plaquette operator $w_p = e^{i\theta_p} + e^{-i\theta_p}$ has norm 2 and shifts the four link charges by ± 1 . Gauss’s law allows decomposition of a nonzero charge network into closed charged cycles. On a periodic cubic lattice with $N_s \geq 4$, every nonempty support has at least four links. A plaquette loop attains this minimum, so the physical electric gap is exactly $2g^2$ in units $\hbar c/a$.

Writing $P = 3N_s^3$, the magnetic operator after removal of its scalar constant has norm at most $2P/g^2$. The two-eigenvalue min–max argument of the corrected Kogut–Susskind note therefore gives

$$\left(2g^2 - \frac{4P}{g^2}\right) \frac{\hbar c}{a} \leq \Delta_{a,L}^{\text{phys}} \leq \left(2g^2 + \frac{4P}{g^2}\right) \frac{\hbar c}{a}, \quad L = N_s a.$$

At fixed N_s , $\Delta_{a,L}^{\text{phys}}/(g^2 \hbar c/a) \rightarrow 2$. The lower bound is extensive and supplies no volume-uniform threshold. A nonpositive lower bound is uninformative.

The former adjacent-growth calculation controls a restricted history class. It misses histories reaching a test link later and does not establish true-vacuum decay; changing the gauge group does not repair those gaps. Uniformity at sufficiently strong coupling follows instead from the group-general Yarotsky application. Its constants depend on the chosen local interaction range.

2. What differs is the weak-side flow

The target-box note now provides an existential strong-coupling region for a specified family of finite-range local corrections. Its edge and tolerance are not the former continuous-time numerical values. An effective Hamiltonian must satisfy its hypotheses before its gap can be transferred.

For $SU(3)$, perturbative asymptotic freedom gives $g^2(2a) = g^2(a) + 2b_0 \log 2 g^4(a) + O(g^6)$ at weak coupling, with $2b_0 \log 2 \simeq 0.0966$. A proposed route is to control all generated corrections until the effective Hamiltonian enters that region. This requires a constructive reduction and

spectral transfer; absence of an intermediate fixed point is not an equivalent proof. The position note §6 identifies the region where neither the perturbative expansion nor a strong-coupling theorem applies.

For continuum pure $U(1)$, perturbative running is absent to all orders. That is not an exact blocking identity for the compact lattice action at every coupling. The weak-coupling massless phase and the existential strong-coupling gap do not alone identify the entire gapped set as one interval or locate its boundary.

3. Non-abelian mechanisms and their limits

The position note §4 isolates the commutator potential of the zero-mode sector, the absence of a gauge-invariant operator linear in the non-abelian electric field, and the one-loop valley potential. These are distinct mechanisms: a small-volume confining potential, a missing photon observable, and a perturbative correction. They motivate the non-abelian route but do not establish the full continuum gap from the beta function alone.

4. What this settles and what it leaves

Both groups have a gap at sufficiently strong coupling by the existential theorem. Compact $U(1)$ also has a massless weak phase. For $SU(3)$ the perturbative slope is positive, but controlled arrival, scaling and continuum reconstruction remain open. The strong-coupling edge and tolerance in this Hamiltonian comparison are existential.

5. Consequence for STATE

Use the group-general existential strong-coupling theorem and the explicit finite-volume bound above. Do not reuse the withdrawn uniform 4λ rate or numerical box location. A non-abelian continuum gap still requires controlled reduction, scaling and reconstruction; perturbative running alone is insufficient. This written correction supplies no new explicit volume-uniform threshold.